(a) Jacobian $J_t = \partial \hat{y}_t / \partial z_t$
1.00
0.75
0.50
0.25
0.00
R^2
20
40
context length t
$R^2(J_t,\ \hat{\theta}_t^{\text{Bayes}})$ (left)
$\|J_t - \theta\|^2$
$\|\hat{\theta}_t^{\text{Bayes}} - \theta\|^2$
10^{-1}
squared error
(b) dimension-matched probes
held-out probe R^2 at $t = 47$
1.0
0.5
0.0
1
2
3
4
block
K,V fast weights (4224-d) → (S, C)
K,V fast weights → $\hat{\theta}$
same, random 128-d projection → $\hat{\theta}$
residual, all 48 positions (6144-d) → $\hat{\theta}$
residual, query position (128-d) → $\hat{\theta}$
(c) causal: patch only K,V
transfer τ
1.0
0.5
0.0
1
2
3
4
3+4
2-4
K 3+4
V 3+4
context K,V patched (blocks)